

Classic Riddler

22 April 2022

Riddle:

In the hit online game World of Riddlcraft, players can level up their armor. Armor levels range from 0 to 5. Now, attempting to level up your armor requires a cerulean gem, which is destroyed in the process. If the attempt is successful, your armor's level goes up by one; if not, it goes down by one.

Fortunately, it's impossible to fail when attempting to upgrade your armor from level 0 to level 1. However, the likelihood of success goes down the higher level the armor is before the upgrade. More specifically:

- Upgrading from level 0 to level 1 has a 100 percent chance of success.
- Upgrading from level 1 to level 2 has an 80 percent chance of success.
- Upgrading from level 2 to level 3 has a 60 percent chance of success.
- Upgrading from level 3 to level 4 has a 40 percent chance of success.
- Upgrading from level 4 to level 5 has a 20 percent chance of success.

On average, how many cerulean gems can you expect to use up in order to upgrade your armor from level 0 to level 5?

Solution:

I wrote the script `gems.py` to simulate this game. With 10^5 iterations, the average number of gems used comes out to approximately 42.8.